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Physics-Informed and Uncertainty-Aware Stacking Ensemble for Robust Prediction of Concrete Compressive Strength

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Received: 13 April 2026 | **Revised:** 13 June 2026 | **Accepted:** 26 June 2026

Academic Editor: Angelo Marcelo Tusset

Keywords: concrete compressive strength | gradient boosting | out-of-fold learning | physics-informed machine learning | stacking ensemble | uncertainty quantification

ABSTRACT

Accurate prediction of concrete compressive strength is essential for structural safety, material optimization, and cost-efficient construction. While machine learning models have demonstrated strong predictive capabilities, many existing approaches rely on overparameterized architectures, lack physical interpretability, and fail to provide uncertainty estimates required for engineering decision-making. This study proposes a physics-informed and uncertainty-aware stacking ensemble framework for robust prediction of concrete compressive strength. The proposed approach integrates domain-informed feature engineering with monotonicity-constrained gradient boosting models to enforce physically consistent relationships between input variables and strength development. A statistically rigorous out-of-fold stacking strategy is employed to generate unbiased meta-features, which are subsequently combined using an ElasticNet meta-learner to improve generalization. To enhance reliability, the framework incorporates quantile regression for uncertainty estimation, enabling prediction intervals alongside point estimates. Experimental evaluation on the benchmark UCI concrete compressive strength dataset comprising 1030 samples with eight input features demonstrates that the proposed framework achieves competitive predictive performance while significantly improving model robustness and interpretability. Ablation analysis confirms that removing redundant feature transformations and enforcing physics-guided constraints enhance generalization. The proposed method provides a reliable and practically applicable solution for data-driven concrete strength prediction, bridging the gap between advanced machine learning and engineering domain requirements.

1 | Introduction

Concrete is the most widely used construction material globally, and its compressive strength is a fundamental parameter governing structural safety and performance. Accurate estimation of compressive strength is critical for optimizing mix design, reducing material costs, and ensuring reliability in construction processes. Traditional laboratory-based testing methods are time-consuming, resource-intensive, and often impractical for rapid decision-making in large-scale projects.

In recent years, machine learning (ML) techniques have emerged as effective alternatives for predicting concrete compressive strength due to their ability to model complex nonlinear relationships among mixture components. Various models, including artificial neural networks, support vector machines, and

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gradient boosting (GB) algorithms, have demonstrated strong predictive performance on benchmark datasets. Ensemble learning methods, particularly stacking-based frameworks, have further improved prediction accuracy by combining multiple learners.

Despite these advancements, several limitations persist in existing approaches. First, many models prioritize predictive accuracy without incorporating domain knowledge, leading to physically inconsistent predictions. Second, conventional stacking methods often suffer from information leakage due to improper meta-feature construction, which compromises generalization. Third, most studies focus solely on point predictions and neglect uncertainty quantification, which is crucial for risk-sensitive engineering applications. Finally, increasing model complexity through feature expansion and hybridization does not necessarily lead to improved performance and may degrade generalization.

To address these challenges, this study proposes a novel ML framework that integrates physics-informed modeling, statistically rigorous ensemble learning, and uncertainty-aware prediction. The proposed method introduces three key contributions.

First, a physics-informed learning strategy is incorporated by enforcing monotonicity constraints on critical variables such as the water-to-cement ratio and curing age, ensuring that model predictions remain consistent with established engineering principles.

Second, a robust out-of-fold (OOF) stacking mechanism is employed to construct unbiased meta-features, significantly improving generalization compared to conventional stacking approaches.

Third, the framework integrates quantile regression to estimate predictive uncertainty, enabling the generation of confidence intervals alongside point predictions, which enhances reliability for practical decision-making.

Unlike existing studies that focus primarily on improving accuracy through increasing architectural complexity, this work emphasizes robustness, interpretability, and reliability. By combining physics-guided constraints, principled ensemble learning, and uncertainty quantification, the proposed framework provides a more comprehensive and practically relevant solution for concrete compressive strength prediction. The dataset employed in this study contains eight input variables and one output variable. The input features comprise quantities (in kg/m^3) of cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, and fine aggregate, along with the curing age (in days). The output variable is the concrete compressive strength measured in megapascals (MPa). A detailed statistical description and the derivation of additional engineered features from these inputs are provided in Section 3.

2 | Literature Review

We have studied several works related to the research for invaluable insights.

Zhang et al. [1] proposed an interpretable ML framework utilizing ensemble algorithms such as RF, adaptive boosting (AdaBoost), XGBoost, and LightGBM, optimized through random search and 5-fold cross-validation (CV) to forecast high-performance concrete (HPC) strength. The methodology relied on a dataset of 1030 samples collected from the UCI database, which was further enhanced through the derivation of six additional feature parameters. A key limitation noted is the inherent nonlinearity of concrete data, which makes establishing simple one-to-one mapping relationships challenging. To improve future models, the authors plan to address data distortion and data drift in real engineering applications by integrating dimensionality reduction and higher-performance deep learning architectures.

Shaaban et al. [2] utilized a supervised regression methodology implemented in Python, investigating various models including linear regression, ridge, lasso, decision trees, random forest, SVR, and KNeighbors regressor, ultimately identifying XGBoost as the top performer through hyperparameter tuning and CV. The study employed a dataset of 167 original experimental records of high-strength concrete, featuring parameters such as cement, silica fume, water, superplasticizer, sand, gravel, and curing age. A primary limitation noted was the model's sensitivity to data distribution, which can lead to difficulties when extrapolating beyond the training range or handling imbalanced datasets. To improve these models, the authors suggested applying advanced feature engineering strategies, expanding the data to include environmental variables such as humidity, and conducting further validation on external datasets.

Palanisamy et al. [3] integrated IoT-based continuous temperature sensors with artificial neural networks (NN-LM and NN-GD) and the maturity method to estimate concrete strength in real-time. The dataset comprised laboratory-tested samples of concrete grades M20–M40 with curing durations ranging from 1 to 28 days. Limitations included a high dependency on sensor reading accuracy and limited generalizability beyond the specific grades and environmental conditions tested. Future improvements involve developing hybrid models that combine the maturity method with ANNs and integrating these systems with digital twin frameworks or BIM platforms for better infrastructure management. The study by Baseer and Ali [4] formulated strength prediction as a supervised regression task, evaluating ensemble algorithms such as random forest, extra trees, and XGBoost, with the gradient boosting regressor (GBR) selected for its robustness and interpretability. The methodology utilized a curated dataset of 253 mix designs from published literature, which was cleaned to 226 observations and supplemented by 18 laboratory-tested validation samples. Limitations included inherent variability from compiling multisource literature data and a narrow focus on 28-day strength, which excluded other mechanical properties. Future improvements could involve expanding the dataset to include binder types and recycled materials, exploring deep learning architectures, and integrating uncertainty quantification techniques such as Bayesian approaches.

Yang et al. [5] evaluated multiple algorithms, including SVR, XGBoost, and BP-ANN, synthesized with the particle swarm optimization (PSO) algorithm to automatically tune hyperparameters for fly ash concrete (FAC) strength prediction. The dataset consisted of 177 preprocessed records from a transportation research institute, focusing on variables such as water-

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to-binder ratios and fly ash replacement rates. A primary limitation is that these models are scenario-specific and may not perform as well when applied to other metrics like carbonization or chloride diffusion. Future work could improve usability by exploring a wider variety of ML algorithms and constructing a specialized operational interface for field applications.

Benaicha [6] employed backpropagation neural networks (BPNN) to handle the nonlinear interactions between ultrasonic pulse velocity (UPV) measurements and concrete compressive strength. The study utilized an extensive dataset of self-compacting concrete (SCC) specimens, incorporating numeric values for hardened density and strength at 1, 7, and 28 days. Limitations centered on the potential for skewed distributions and outliers to negatively impact predictive analysis if not proactively addressed. The authors suggested that further research into the interplay between air-entraining agent (AEA) dosages and concrete performance is necessary to improve durability predictions.

Zhang et al. [7] developed a hybrid PCA–PSO–RVM collaborative optimization model that uses principal component analysis for dimensionality reduction and PSO to determine the optimal hyperparameters for a relevance vector machine. The dataset comprised 99 experimental samples of SCC extracted from existing literature, focusing on components such as cement, sand, fibers, and admixtures. Despite its high accuracy, the model suffers from limited interpretability, making it difficult for engineers to intuitively understand the impact of specific features on the final prediction. Suggested improvements include using data enhancement technologies to boost training efficiency and developing more visualization-friendly models to increase acceptability in practical engineering applications.

Fathy et al. [8] utilized XGB, M5P-tree, and RF models combined with grid search optimization and Shapley additive explanations (SHAP) interpretability analysis to predict the strength of concrete containing waste powders. The methodology was applied to an experimental dataset of 324 records involving granite and marble waste powders subjected to elevated temperatures up to 800°C. Limitations included a restricted range of waste material doses (0%–9%) and a narrow focus on compressive strength. Suggested improvements include incorporating more input parameters and expanding the framework to predict other mechanical properties such as flexural and tensile strength.

Mahmood et al. [9] compared traditional ML models with BPNN using various optimizers (Adam, SGD, and RMSprop) to forecast the strength of SCC modified with sustainable replacements. The experimental dataset included 133 samples using rice husk ash (RHA) as a cement replacement and marble powder (MP) as a filler. A significant limitation was the poor performance of individual linear ML models, which failed to capture the nonlinear complexities of the data. Future research could improve outcomes by focusing on the unique synergistic effects of ternary material blends and expanding dataset sizes for higher accuracy.

Alghrairi et al. [10] employed the KNIME platform to compare nine different algorithm strategies, including gradient boosted trees (GBTs) and RF, to predict the strength of lightweight concrete (LWC). The study drew from a massive dataset of 2568 samples extracted from 152 peer-reviewed academic publications. Limitations involve a limited understanding of the underlying causality behind scientific principles and the challenge of obtaining consistently high-quality, representative data. Future work should focus on collecting even larger unified databases and developing empirical models for mechanical properties like the modulus of elasticity.

The research by Chakma et al. [11] evaluated three hybrid frameworks: ET–XGB, RF–LGBM, and Transformer–XGB—utilizing SHAP analysis and uncertainty quantification to predict compressive, flexural, and tensile strengths. The dataset was extensive, containing 703 compressive strength records and hundreds of flexural and tensile strength samples compiled from multiple experimental studies. A significant limitation was the high predictive variability and uncertainty found in the Transformer–XGB model, suggesting it may struggle with generalization compared to the more stable ET–XGB. The study recommends improving reliability by selecting the specific hybrid model that matches the target property (e.g., ET–XGB for compressive strength) and focusing mix design adjustments on the most influential features identified by SHAP, such as fiber aspect ratios.

Nugraha et al. [12] introduced a cooperative neural network (CoNN) framework that pairs an autoencoder-based imputation model with a surrogate strength predictor to perform constraint-aware mix generation in a single forward pass. The researchers utilized a benchmark dataset of 1030 HPC records (filtered to 749 samples for curing ages up to 28 days), focusing on variables such as cement, slag, fly ash, and superplasticizer. A limitation of the study is the relatively scarce dataset, which might impact the model's broader generalizability across more diverse engineering scenarios. Future work could improve the framework by incorporating larger-scale datasets and exploring physics-informed neural networks that inject mechanistic domain knowledge directly into the learning process.

Pan et al. [13] employed a stacking ensemble model that integrated XGBoost, extra trees, and decision trees as base learners with a linear regression meta-learner, optimized via random search and 10-fold CV. The methodology was applied to a combined dataset of 63 self-made recycled concrete records and 1030 samples from the UCI repository. One identified limitation is that pursuing extremely high statistical metrics can occasionally lead to overfitting, particularly when compared to simpler integration strategies. To improve future iterations, the authors suggest using feature expansion and extraction to simplify input combinations and focusing on high-impact variables like the water-to-cementitious ratio identified through SHAP analysis.

Kashem et al. [14] developed hybrid ensemble models (AB-PSO, RF-PSO, and GB-PSO) using PSO for hyperparameter tuning and SHAP/PDP for interpretability. The dataset comprised 810 data points extracted from literature covering various ultra-high-performance concrete (UHPC) mixtures. One limitation identified was the variability in prediction uncertainty between different hybrid architectures, as shown in uncertainty analysis. Future improvements could involve using gray wolf optimization (GWO) or hybrid deep learning models like LSTM and CNN to further refine accuracy.

Naved et al. [15] focused on deep neural network (DNN) models optimized through the Keras Tuner library to identify the best

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configurations for hidden layers, neurons, and learning rates. The models were trained and tested on a standard experimental dataset of 1030 instances containing ingredients and curing age as input features. A key limitation discussed was that traditional manual tuning is often suboptimal and inefficient, and the model showed slightly lower performance when predicting higher strength values. Improvements can be achieved by utilizing the tanh activation function over ReLU for better accuracy and leveraging future advancements in automated hyperparameter tuning technologies.

Kumar et al. [16] investigated hybrid models such as GB-differential evolution (GB-DE) and GB-PSO to predict the strength of concrete with high-volume fly ash. The study focused on high-volume fly ash and silica fume blends to reduce carbon emissions from cement production. A major limitation was the relatively small sample size of the experimental data, which emphasizes the need for larger verification sets. To improve reliability, the authors recommend benchmarking against larger datasets and including more diverse cementitious materials.

Sun et al. [17] introduced a Whale Optimization Algorithm–optimized XGBoost (WOA-XGBoost) framework, systematically comparing it against linear regression and RF models. The analysis utilized a high-quality dataset of 1030 mix proportion sets to capture the evolution of concrete strength under multifactor coupling. An identified limitation was the relative lack of systematic studies on the synergistic effects of multiple additives beyond single-additive systems. Future work aims to provide more reliable modeling tools for complex industrial and agricultural waste concrete and identify optimal ranges for key parameters.

Lei et al. proposed a ML framework for predicting HPC strength and visualizing feature contributions to better interpret the influence of mixture components on compressive strength development [18].

Elshaarawy et al. [19] developed a hybrid framework using five GBMs (SGB, XGB, LGB, CGB, and NGB) optimized via Bayesian optimization for simultaneous strength classification and prediction. The dataset was a diverse compilation of 3415 individual samples, processed for outliers and normalized to ensure robustness. A limitation is that the current model is restricted to conventional concrete and does not yet account for ultra-high-performance or fiber-reinforced types. Future research should incorporate durability and sustainability factors and explore the integration of physics-informed hybrid models to better align with engineering principles.

Abdullah et al. [20] employed GB and AdaBoost to predict the unconfined compressive strength (UCS) of soil, comparing their performance against models like ANN and random forest. The methodology used a database of 270 geopolymer-stabilized soil samples, with features including slag and fly ash content, molarity, and various chemical ratios. A primary limitation is that these models are only validated within the specific ranges of the current dataset, necessitating further verification for different soil types or binder ranges. To improve generalizability, the authors suggest collecting more diverse experimental data and exploring sophisticated deep learning algorithms for future prediction tasks.

Falah et al. [21] developed a deep learning neural network (DLNN) coupled with the XGBoost algorithm for feature selection, comparing its efficacy against MARS, ELM, and RF models. The methodology utilized 353 experimental samples of environmentally friendly concrete and tested three different data split scenarios to determine the impact of training size on accuracy. A limitation noted was that standalone AI models often exhibit lower predictability than coupled hybrid systems, potentially leading to underfitting if training data are insufficient. Future improvements could involve introducing even more advanced hybrid architectures and applying uncertainty analysis to better understand model sensitivity and input variable efficiency.

Several recent studies have further broadened the applicability of ensemble and physics-informed learning in engineering contexts. Ensemble regression frameworks with uncertainty-aware outputs have been applied to structural health monitoring and predictive maintenance [22, 23]. Physics-informed hybrid models incorporating domain constraints have demonstrated improvements in dental material property prediction [24] and smart manufacturing quality control [25]. In the context of built environment modeling, recent work has explored uncertainty-quantified ML for building performance and energy estimation [26]. These developments reinforce the relevance of the proposed framework's emphasis on reliability and interpretability alongside predictive accuracy.

3 | Dataset Description

The dataset [27] that was utilized in this research is based on experimental concrete mix records that contain data regarding the composition of the mixture and the compressive strength data of same. There are 1030 samples in the dataset where there are eight inputs and one target variable.

The input variables are the amount of various concrete components in kilograms per cubic meter. Such variables are cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and curing age. The dependent variable will be compressive strength of concrete expressed in MPa.

Observations are associated with their own mixture structure with varying compositions of materials and curing conditions. The large range of mix composition enables the dataset to reflect the different concrete formulations practiced in real-life construction situations.

Some of the derived features based on domain knowledge in concrete mix design were also created to enhance model learning. These characteristics are binder content, water-to-binder ratio, water-to-cement ratio, superplasticizer ratio, aggregate ratio, and logarithmic transformation of curing age.

The dataset was split into training and testing subsets using an 80:20 ratio. Standard normalization was used to perform feature scaling to ensure that there are consistent distributions of features, and a ML algorithm is more stable in its learning.

4 | Methodology

4.1 | Overview

This study proposes a physics-informed and uncertainty-aware stacking ensemble framework for predicting concrete

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compressive strength. The framework integrates domain knowledge, robust ensemble learning, and uncertainty estimation to improve predictive reliability and generalization.

The pipeline consists of the following stages: (i) domain-informed feature engineering, (ii) physics-constrained base learners, (iii) OOF meta-feature generation, (iv) stacking-based meta-learning, and (v) uncertainty quantification.

4.2 | Domain-Informed Feature Engineering

Let the original feature vector be defined as

$$X = [C, S, F, W, SP, CA, FA, \text{Age}]. \tag{1}$$

Derived features are constructed using established principles of concrete mix design:

$$B = C + S + F, \tag{2}$$

$$w_b = \frac{W}{B}, w_c = \frac{W}{C}, \tag{3}$$

$$r_{\text{agg}} = \frac{CA}{FA}, \tag{4}$$

$$\text{Age}_{\log} = \log(1 + \text{Age}). \tag{5}$$

These features improve representational capacity while maintaining physical interpretability.

Design decision: Polynomial interaction features are excluded due to their negative impact on tree-based models, as demonstrated in the ablation study.

4.3 | Physics-Informed GB Models

Three GB models are used as base learners:

- XGBoost
- LightGBM
- CatBoost

To ensure physical consistency, monotonicity constraints are imposed:

$$\frac{\partial \hat{y}}{\partial (w/c)} < 0, \frac{\partial \hat{y}}{\partial \text{Age}} > 0. \tag{6}$$

These monotonicity constraints are explicitly enforced in GB frameworks that support constrained optimization, such as XGBoost and LightGBM.

For XGBoost and LightGBM, the monotonicity constraints in Equation (6) are enforced natively via their built-in monotone constraint parameters. Specifically, the water-to-cement ratio feature (w/c) is assigned a constraint value of -1 (enforcing a strictly nonincreasing relationship with predicted strength), and the curing age feature (Age) is assigned a constraint value of $+1$ (enforcing a strictly nondecreasing relationship). These settings guarantee that, for any two samples differing only in w/c or Age, the model's output will never violate the specified monotonic direction across any leaf split in any tree.

CatBoost does not expose an equivalent monotone constraint API for regression tasks in the version used in this study. Approximate monotonic consistency for CatBoost is achieved through two measures. First, the water-to-binder ratio (w/b) and water-to-cement ratio (w/c) are included as explicit engineered features so that the model can represent these physically critical relationships directly, reducing the probability that internal tree splits produce inversions. Second, strong L2 regularization (via the l2_leaf_reg hyperparameter, tuned via Bayesian optimization) penalizes unstable leaf values, smoothing the response surface and suppressing monotonicity violations in high-variance regions. To verify approximate compliance post-training, the CatBoost model's predictions are evaluated on a held-out monotonicity test set constructed by independently varying w/c across its observed range while holding all other features at their median values; the proportion of adjacent prediction pairs that satisfy $\hat{y}(w/c_i) \geq \hat{y}(w/c_{i+1})$ for $w/c_i < w/c_{i+1}$ is reported as the monotonicity compliance rate. In our experiments, this rate exceeded 97 % , confirming adequate approximate enforcement.

This ensures that all base learners adhere, either strictly or approximately, to known physical relationships.

The objective function is defined as

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k} \Omega(f_k). \tag{7}$$

4.4 | OOF Meta-Feature Generation

To avoid information leakage, K-fold OOF stacking is employed.

Let f_1, f_2, f_3 denote base learners. Meta-features are generated as

$$Z = [z_1, z_2, z_3], \tag{8}$$

where each z_j is obtained using OOF predictions.

4.5 | Stacking-Based Meta-Learning

The final prediction is obtained using a meta-learner:

$$\hat{y} = g(Z). \tag{9}$$

ElasticNet regression is used:

$$\min_{w} \left(\|y - Zw\|_2^2 + \alpha \left(\lambda \|w\|_1 + (1 - \lambda) \|w\|_2^2 \right) \right). \tag{10}$$

4.6 | Uncertainty Quantification via Quantile Regression

To estimate predictive uncertainty, quantile regression is applied as a parallel branch operating on the stacking meta-features Z, not on the original feature set X. Specifically, after the OOF meta-features are constructed (Section 4.4), three separate GB quantile regressors are trained on Z to predict the $\tau = 0.1$, $\tau = 0.5$, and $\tau = 0.9$ conditional quantiles:

$$\hat{y}_\tau = f_\tau(Z), \quad \tau \in \{0.1, 0.5, 0.9\}. \tag{11}$$

This design ensures that the uncertainty model operates on the same compressed, unbiased representation used by the point-prediction meta-learner, thereby maintaining consistency between the two outputs. The $\tau = 0.5$ quantile model provides an

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alternative median estimate, while the $\tau = 0.1$ and $\tau = 0.9$ models define the lower and upper bounds of the 80% prediction interval:

$$\left[\widehat{y}_{0.1}, \widehat{y}_{0.9}\right]. \tag{12}$$

The quantile models are trained using the asymmetric pinball loss:

$$\mathcal{L}_\tau(y, \widehat{y}) = \begin{cases} \tau(y - \widehat{y}), & y \geq \widehat{y}, \\ (1 - \tau)(\widehat{y} - y), & y < \widehat{y}. \end{cases} \tag{13}$$

Crucially, the quantile regression branch is trained and evaluated on the held-out test fold exclusively, using the same OOF meta-features already generated for the point-prediction model. No additional data leakage is introduced. The point-prediction pipeline (ElasticNet meta-learner) and the uncertainty pipeline (quantile regressors on Z) are therefore structurally parallel, sharing the same meta-feature input but optimizing different objectives.

4.7 | Hyperparameter Optimization

Hyperparameters are optimized using Bayesian optimization:

$$\theta^* = \arg\max_\theta \mathbb{E}_{CV}\left[R^2(\theta)\right]. \tag{14}$$

4.8 | Computational Complexity

The overall complexity is

$$\mathcal{O}(K \cdot M \cdot T \cdot \log n), \tag{15}$$

where K is the number of folds, M is the number of models, and T is the number of trees.

5 | Results and Evaluation

5.1 | Experimental Setup

The proposed framework is evaluated on the UCI Concrete Compressive Strength dataset, consisting of 1030 samples with eight input variables and one target variable. The dataset is split using an 80:20 train–test ratio. In addition, a 10-fold *CV* strategy is employed to ensure robust performance estimation and reduce variance. To avoid data leakage, the 10-fold CV is performed exclusively on the training set, while the held-out test set remains completely unseen and is used only for final performance evaluation.

All features are standardized prior to model training to ensure stable optimization of the ElasticNet meta-learner. While feature scaling is not required for tree-based base learners, it is applied for consistency within the stacking framework. Model performance is evaluated using the following metrics:

- Coefficient of determination (R^2)
- Mean absolute error (MAE)
- Root mean squared error (RMSE)

All experiments are conducted with fixed random seeds (seed = 42) to ensure reproducibility. The implementation is carried out using Python with libraries including NumPy, Pandas, Scikit-learn, XGBoost, LightGBM, and CatBoost. Experiments are executed on a machine with an Intel Core i5 processor, 16 GB RAM, and no GPU acceleration.

5.2 | Comparison With Baseline Models

Table 1 compares the proposed method with standard ML models.

The proposed framework outperforms all author-implemented baseline models under identical experimental conditions (Table 1), demonstrating the effectiveness of combining physics-informed learning with robust stacking.

It is acknowledged that certain published state-of-the-art implementations on the same UCI benchmark achieve higher raw accuracy (Table 2); the primary contribution of the proposed method, however, lies in its combination of reliability, physical interpretability, and uncertainty quantification rather than in optimizing predictive accuracy alone.

5.3 | Comparison With State-of-the-Art Methods

To assess the competitiveness of the proposed model, its performance is compared with recent studies using the same benchmark dataset.

Table 2 shows that the proposed model ($R^2 = 0.945$) does not surpass the best published methods on the UCI benchmark in terms of raw accuracy as Zhang et al. [1] achieve $R^2 = 0.966$ and Kumar et al. [28] achieve $R^2 = 0.961$. This gap is acknowledged. The proposed framework is not primarily designed to maximize predictive accuracy; rather, it addresses three limitations that those methods do not simultaneously resolve:

- Physically consistent predictions via monotonic constraints
- Robust generalization through OOF stacking
- Uncertainty estimation for risk-aware decision-making

This highlights that the contribution of this work extends beyond raw accuracy to include reliability and interpretability.

It is important to note that the CatBoost performance reported in Table 1 ($R^2 = 0.928$) represents our own reimplementation of a standalone CatBoost baseline trained on the standard 80:20 split without additional feature engineering, hyperparameter tuning beyond default settings, or dataset augmentation. In contrast, the CatBoost result of Zhang et al. [1] ($R^2 = 0.966$) was obtained using an extended feature set derived from 1030 UCI samples enriched with six additional domain-specific features and optimized through random search with 5-fold CV. The two figures are therefore not directly comparable: Table 1

TABLE 1 | Performance comparison with baseline models.

Model	MAE (MPa)	RMSE (MPa)	R^2
Linear regression	5.214	7.102	0.812
Random forest	3.102	4.892	0.901
XGBoost	2.996	4.606	0.918
LightGBM	2.873	4.421	0.923
CatBoost	2.812	4.302	0.928
Proposed model	**2.601**	**3.842**	**0.945**

Note: The bold values indicate the outcomes from the proposed approach.

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TABLE 2 | Comparison with state-of-the-art models on the UCI concrete dataset.

Study	Model type	MAE (MPa)	RMSE (MPa)	R^2
Zhang et al. [1]	CatBoost	2.27	3.06	0.966
Kumar et al. [28]	Optimized LightGBM	2.446	3.371	0.961
Proposed Model	Physics-informed stacking	**2.601**	**3.842**	**0.945**

Note: The bold values indicate the outcomes from the proposed approach.

benchmarks model families under identical conditions to isolate the effect of the proposed stacking framework, while Table 2 reflects the best published configurations available in the literature on the same benchmark dataset.

5.4 | Uncertainty Evaluation

The proposed model generates prediction intervals using quantile regression. The quality of uncertainty estimation is evaluated using prediction interval coverage probability (PICP) and mean prediction interval width (MPIW).

While the proposed framework incorporates quantile regression for uncertainty estimation, direct comparison with alternative uncertainty modeling approaches such as quantile random forests or Bayesian regression methods is left for future work. The focus of this study is to demonstrate the integration of uncertainty estimation within a physics-informed stacking framework rather than benchmarking uncertainty techniques (Tables 3 and 4).

The results indicate that the predicted intervals reliably capture the true values while maintaining reasonable tightness, making the model suitable for engineering applications.

5.5 | Statistical Significance Analysis

To validate the performance improvement, a paired t-test is conducted between the proposed model and the best baseline (CatBoost). The null hypothesis assumes no significant difference in prediction errors.

The test results yield $p < 0.01$, indicating that the improvement achieved by the proposed framework is statistically significant.

5.6 | Discussion

The experimental results demonstrate that the proposed framework achieves strong predictive performance while addressing key limitations of existing models. Unlike purely accuracy-driven approaches, the proposed method balances performance, interpretability, and reliability, making it suitable for real-world engineering applications.

TABLE 3 | Uncertainty estimation performance.

Metric	Value
PICP (90% interval)	0.914
MPIW (MPa)	5.12

6 | Ablation Study

6.1 | Objective

The ablation study is conducted to evaluate the contribution of each component of the proposed framework. Specifically, the impact of feature engineering, polynomial features, physics constraints, stacking strategy, and uncertainty modeling is analyzed.

6.2 | Experimental Configurations

The following configurations are evaluated:

- Baseline (XGBoost with raw features)
- + Domain-informed features
- + Polynomial features
- + Hyperparameter optimization
- + Physics constraints
- + OOF stacking (proposed core)
- + Uncertainty modeling (full model)

7 | Results

7.1 | Key Findings

Effect of feature engineering: Domain-informed features provide consistent improvement by incorporating physically meaningful relationships.

Effect of polynomial features: The addition of polynomial interaction features significantly degrades performance. This confirms that tree-based ensemble models inherently capture nonlinear interactions, making explicit polynomial expansion redundant and harmful.

Effect of hyperparameter optimization: Hyperparameter tuning provides marginal improvements, indicating that model architecture plays a more critical role than fine-tuning alone.

Effect of physics constraints: Incorporating monotonic constraints improves both accuracy and interpretability, demonstrating the importance of embedding domain knowledge.

Effect of OOF stacking: OOF stacking yields the largest performance gain, highlighting the importance of unbiased meta-learning.

Effect of uncertainty modeling: The inclusion of uncertainty modeling does not affect point prediction metrics such as MAE, RMSE, and R^2, as it is designed to estimate prediction intervals rather than improve deterministic accuracy. However, it significantly enhances model reliability by providing confidence

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TABLE 4 | Ablation study results.

Model variant	MAE (MPa)	RMSE (MPa)	R^2
Baseline (XGBoost)	2.996	4.606	0.918
+ Feature engineering	2.882	4.421	0.923
+ Polynomial features	3.317	5.238	0.889
+ Hyperparameter optimization	2.910	4.512	0.920
+ Physics constraints	2.781	4.203	0.932
+ OOF stacking	2.601	3.842	0.945
+ Uncertainty modeling (full model)	2.601	3.842	0.945

bounds, which are essential for risk-sensitive engineering applications.

7.2 | Discussion

The ablation study reveals that increasing model complexity does not necessarily lead to better performance. Instead, principled design choices—such as removing redundant features, enforcing physical constraints, and adopting robust stacking strategies—play a more significant role in improving generalization.

These findings directly address common issues in existing literature, where overly complex pipelines often fail to deliver proportional performance gains.

8 | Future Work

Future research can extend the proposed framework in several directions. First, incorporating large-scale industrial datasets with environmental and operational variables would improve generalization and real-world applicability. Second, physics-informed neural networks and graph-based learning approaches may further enhance the modeling of complex material interactions. Third, integrating multitask learning could enable simultaneous prediction of multiple concrete properties, such as tensile strength and durability. Additionally, uncertainty estimation can be further refined using Bayesian deep learning techniques to provide more reliable confidence measures. Finally, the integration of explainable AI methods such as SHAP and counterfactual analysis could improve interpretability and support engineering decision-making.

9 | Conclusion

This study presents a physics-informed and uncertainty-aware stacking ensemble framework for predicting concrete compressive strength. The proposed approach integrates domain-informed feature engineering, monotonicity-constrained GB models, OOF stacking, and quantile regression to improve predictive reliability and generalization.

Experimental results demonstrate that the framework achieves strong predictive performance while ensuring physical consistency and providing uncertainty estimates. The ablation study highlights that removing redundant feature expansions and adopting principled ensemble learning significantly improve model robustness.

Unlike conventional approaches that prioritize accuracy alone, this work emphasizes reliability, interpretability, and practical applicability. The proposed framework provides a robust and scalable solution for data-driven modeling in civil engineering and offers a foundation for future research in physics-informed ML for material prediction tasks.

Nomenclature

C	Cement content (kg/m^3)
S	Blast furnace slag content (kg/m^3)
F	Fly ash content (kg/m^3)
W	Water content (kg/m^3)
SP	Superplasticizer content (kg/m^3)
CA	Coarse aggregate content (kg/m^3)
FA	Fine aggregate content (kg/m^3)
Age	Curing age (days)
B	Total binder content ($C + S + F$)
w/b	Water-to-binder ratio
w/c	Water-to-cement ratio
r_{agg}	Aggregate ratio (CA/FA)
$\text{Age}_{\log}$	Log-transformed curing age
$\hat{y}$	Predicted compressive strength (MPa)
$\hat{y}_\tau$	τ-quantile prediction
$\mathcal{L}_\tau$	Quantile (pinball) loss function
K	Number of cross-validation folds
M	Number of base learner models
T	Number of trees per model
n	Number of training samples
Z	Meta-feature matrix from OOF predictions
α, λ	ElasticNet regularization hyperparameters
ML	Machine learning
OOF	Out-of-fold
PICP	Prediction interval coverage probability
MPIW	Mean prediction interval width
SHAP	Shapley additive explanations
HPC	High-performance concrete
UCI	University of California, Irvine

Author Contributions

As the sole author, Md. Siam Ansary has conducted the research on his own.

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Funding

The author received no funding for the research work.

Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

Data supporting the findings of this study are available on the UC Irvine Machine Learning Repository.

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